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@
2 #U13157006 李恩頔
3 @ Assembler program to convert a string to
4 @ all lower case. (Exercise 6-3)
5 @
6 @ R0 - string parameter to printf
7 @ R3 - address of output string
8 @ R4 - address of input string
9 @ R5 - current character being processed
10 @
11
12 .thumb_func          @ Necessary because sdk uses BLX
13 .global main          @ Provide program starting address to linker
14
15 main: BL stdio_init_all @ Initialize uart or usb
16
17     LDR R4, =instr      @ start of input string
18     LDR R3, =outstr     @ address of output string
19
20 @ The loop is until byte pointed to by R4 is non-zero
21 loop: LDRB R5, [R4]     @ load character
22     ADD R4, #1          @ increment pointer
23
24 @ If R5 > 'Z' then goto cont
25     CMP R5, #'Z'        @ is letter > 'Z'?
26     BGT cont
27
28 @ Else if R5 < 'A' then goto cont
29     CMP R5, #'A'
30     BLT cont             @ goto to end if
31
32 @ if we got here then the letter is uppercase, so convert it.
33     ADD R5, #('a'-'A') @ 改為加上差值 (轉小寫)
34
35 cont: @ end if
36     STRB R5, [R3]        @ store character to output str
37     ADD R3, #1           @ increment pointer
38     CMP R5, #0           @ stop on hitting a null character
39     BNE loop             @ loop if character isn't null
40
41 @ Setup the parameters to printf our lower case string
42 loop2: LDR R0, =outstr   @ string to print
43     BL printf            @ Call printf to output
44     B loop2
45
46 .data
47 instr: .asciz "This is our Test String that we will convert.\n"
48 outstr: .fill 255, 1, 0
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